Laplace Transform

Monday, December 15, 2025 6:06 PM

Laplace Transform: $$\mathcal{L}[f](s) = \int_0^\infty e^{-st} f(t)\,dt = F(s)$$

① Linearity: $\mathcal{L}[\alpha f + bg](s) = \alpha \mathcal{L}[f] + b\mathcal{L}[g]$

② $\mathcal{L}[e^{ct} g(t)] = F(s-c)$

③ $\mathcal{L}[p^n(f)] = s^n \mathcal{L}[p(t)] - s^{n-1} p(0) - \dots - s p^{(n-2)}(0) - p^{(n-1)}(0)$

Inverse Laplace

Linear: $Q(s) = (s-s_0)(s-s_1)\dots(s-s_n) \Rightarrow F(s) = \frac{a_1}{s-s_0} + \frac{a_2}{s-s_1} + \dots + \frac{a_n}{s-s_n}$

$\uparrow$ $a_1 \mathcal{L}^{-1}[s-s_0] = a_1 e^{s_0 t}$

Repeated linear: $Q(s) = \dots (s-s_i)^k \dots \Rightarrow F(s) = \dots \left(\frac{a_{i1}}{s-s_i} + \frac{a_{i2}}{(s-s_i)^2} + \dots + \frac{a_{ik}}{(s-s_i)^k} \right) + \dots$

$\frac{a_{i1}}{(k-1)!} \mathcal{L}^{-1}\left[\frac{(k-1)!}{(s-s_i)^k}\right] = \frac{a_{i1}}{(k-1)!} t^{k-1} e^{s_i t}$ $\uparrow$

Dirac Delta: $\delta(t) = \begin{cases} 0, & t \neq 0 \\ \infty, & t = 0 \end{cases}$

property: $$\int_{-\infty}^{\infty} f(t)\,\delta(t-a)\,dt = f(a)$$

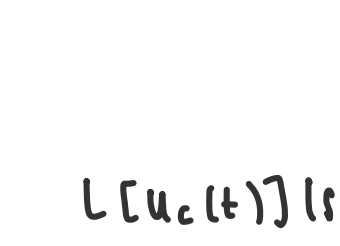
Heaviside: $u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$ $\quad u_c(t) = \begin{cases} 1, & t \geq c \\ 0, & t < c \end{cases}$

$$\begin{aligned} \mathcal{L}[u(t)](s) &= \int_{-\infty}^{\infty} e^{-st} u(t)\,dt \\ &= \int_0^\infty e^{-st}\,dt \\ &= \frac{1}{s} \end{aligned}$$

$$\begin{aligned} \mathcal{L}[u_c(t)](s) &= \int_c^\infty e^{-st}\,dt \\ &= \left[-\frac{1}{s} e^{-st}\right]_c^\infty \\ &= \frac{e^{-sc}}{s} \end{aligned}$$

$$\mathcal{L}[u_c(t) f(t-c)](s) = e^{-sc} \underbrace{F(s)}_{\mathcal{L}[f(t)]}$$

Indicator: $1_A(\vec{x}) = \begin{cases} 1, & \vec{x} \in A \\ 0, & \vec{x} \notin A \end{cases}$



$$\begin{aligned} \mathcal{L}[u_{cd}(t)](s) &= \int_0^\infty e^{-st} u_{cd}(t)\,dt \\ &= \int_c^d e^{-st}\,dt \\ &= \frac{e^{-sc} - e^{-sd}}{s} \end{aligned}$$

Periodic: $f(t) = f(t+T) \Rightarrow$ define $f_T(t) = f(t)[1 - u_T(t)]$

$$f_n(t) = \sum_{k=0}^{n-1} f(t-kT) u_{kT}\,dt$$

$$\lim_{n\to\infty} \mathcal{L}[f_{nT}](s) = \underbrace{F_T(s)}_{\text{only need Laplace of first period}} \frac{1}{1-e^{-sT}}$$